Alec Cheng

Atlanta GA 30318 ● acheng324@gatech.edu ● 760-822-2626 ● alec-y-cheng.github.io

Master's student in Computer Science interested in applying machine learning methods to simulate, predict, and respond to challenges in the built environment. Background in spatial analysis, architecture, and urban planning, with experience using data-driven methods to inform engineering and design decisions.

EDUCATION AND AWARDS

Georgia Institute of Technology, Atlanta, GA — *Aug 2025 — May 2027*

M.S. in Computer Science – Machine Learning Specialization — GPA: 4.00/4.00

Vanderbilt University, Nashville, TN — *Aug 2021 — May 2024*

B.S. in Cognitive Science, B.S. in Architecture and the Built Environment — GPA: 3.97/4.00

Outstanding Graduate in Architecture, Peabody Scholar, Magna Cum Laude

TECHNICAL PROJECTS

ImmerseGT 2026 Hackathon Winner — *April 2026*

- Designed an extended reality application that uses crowdsourcing to document local histories and machine learning models to reconstruct ruins and artifacts
- Won 1st place in TanitXR track, focusing on preservation of local communities and their narratives

3D Scene Reconstruction Project – Machine Learning (CS7641) — *Fall 2025*

- Built a 3D-scene reconstruction pipeline combining transformer-based image classification models and structure-from-motion techniques to generate accurate 3D scene representations from unordered 2D images.

A 2.5D Approach to Training Robust 3D Pose Estimators – Deep Learning (CS7643) — *Spring 2026*

- Built a hybrid 2D-3D pipeline to strengthen the generalizability of 3D human pose estimation models by including domain-specific 2D data in the training of models.

Soccer Agent RL System – Deep Reinforcement Learning (CS8803-DRL) — *Spring 2026*

- Trained an agent to compete in a multi-agent soccer environment using Proximal Policy Optimization while optimizing hyperparameters and reward design.

Campus Pedestrian Simulation – Modeling and Simulations (CSE-6730) — *Spring 2026*

- Modeled pedestrian movement and interaction patterns on the university campus using ODE/PDE-based systems to predict how large-scale school events would impact campus traffic.

EXPERIENCE

Sustainable Urban Systems Lab, Atlanta, GA — *Jan 2026 –*

RESEARCH ASSISTANT

- Implemented a diffusion-based surrogate for fluid flow simulation that reduced the traditional runtime from multiple days to just 0.6 seconds with 94% accuracy.
- Built and optimized training pipelines for physics-informed neural networks and conducted inference over thousands of training images to compare models for an ongoing paper on surrogate ML models in fluid flow.

Barge Design Solutions, Nashville, TN — *May 2024 – Aug 2024*

PLANNING INTERN

- Conducted geospatial and predictive demographic analysis on a large Alabama city that identified optimal Bus Rapid Transit corridor locations, directly informing where to focus federal transportation funds.
- Led the design of a comprehensive plan for a Florida city, communicated design visions to clients of a Nashville mixed-use development through detailed booklets and effective data visualization techniques.
- Explored applications of generative/predictive AI tools in engineering workflows, and scripted ArcGIS workflows.

Vanderbilt Institute for Spatial Research, Nashville, TN — *Oct 2023 – June 2024*

RESEARCH ASSISTANT

- Georeferenced historical site plans onto satellite imagery, revealing hidden boundaries and burial sites underneath Andrew Jackson's Hermitage, published as interactive visualizations to educate local communities.

SKILLS

Programming and ML Frameworks — Python (PyTorch), C, C++, Java, Git, System Design

GIS, Data Visualization, 3D Modeling — ArcGIS/QGIS, Rhino, Blender, Adobe Creative Suite